Adam Shanti

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Education

Virginia Tech — Graduated 2025
B.S. Aerospace Engineering, ***Magna Cum Laude*** — *Blacksburg, VA*

Experience

F/A-18 EFG Systems/Integration Engineer – Active Secret Clearance — August 2025 – Current
Sabre Systems — *NAS Patuxent River, MD*

- Support EFG Air Vehicle platform-level integration across all major engineering groups via **requirements/configuration control, and technical analysis** to drive hardware readiness.
- Serve as **Deputy to the Lead Systems/Integration Engineer**, coordinating cross-functional progression with engineering, maintenance, and test stakeholders for sustainment and upgrades.
- Drive special-topic issue resolution by personal analysis, Subject Matter Expert input, reviewing drawings and technical data, and delivering **risk-informed courses of action** to leadership.
- Support engineering change activity (ECP/EDRAP artifacts), ensuring clear documentation, traceability, and disciplined release of configuration updates.
- Interface with Integrated Product Teams (major engineering groups) to align integration actions with **test readiness** and safe execution of flight and ground test activities.

Drone Park Intern — June 2024 – February 2025
Virginia Tech — *Blacksburg, VA*

- Executed hands-on build and flight-test iterations on my Passion Project, a modular fixed-wing UAV, troubleshooting **propulsion integration** (motor, ESC, power wiring), structure, aerodynamics and control issues (servos, CG, stability).
- Built and improved test methods including **test procedures**, pre-flight checklists, and data logging to increase repeatability and flight readiness.
- Designed CAD parts and field-repair solutions, applying **DFM** thinking to simplify assembly, reduce rework, and speed iteration cycles.

Projects — https://adamshanti.com

Chief Engineer – QF-01 Interceptor (1st Place, Air Vehicle Design) — August 2024 – May 2025
Virginia Tech – AIAA Homeland Defense Interceptor Team — *Blacksburg, VA*

- Led a **10-person multidisciplinary team** through **design-to-test execution** (for LEX), coordinating aerodynamics, propulsion, structures, and controls while driving decisions to closure under aggressive timelines.
- Owned CAD integration and packaging (fuselage, ducts, structure), translating high-level concepts into detailed geometry and supporting **propulsion and aerodynamic design**, including shock inlet analysis and leading-edge extension design and testing.
- Resolved **cross-discipline integration conflicts** using trade studies and clear technical communication to support gated reviews and final program deliverables.

Mk4 Fixed Wing Drone Platform — June 2022 – Present
Virginia Tech Drone Park — *Blacksburg, VA*

- Engineered a custom **supercapacitor power system** and charging architecture, achieving a **1:1 charge-to-flight time** ratio validated through repeated flight tests.
- Designed a modular rib-and-spar wing, carbon-fiber reinforcements, V-tail assembly, and rail-snap fuselage interface for rapid configuration changes and test iteration.
- Fabricated a **3-phase, 6-stator hand-cranked dynamo** for field charging, optimizing electrical output through mechanical design and experimental validation.

Leading Edge Extension (LEX) Flow Visualization Experiment — January 2025 – May 2025
Virginia Tech Open Jet Wind Tunnel — *Blacksburg, VA*

- Designed and executed a **wind tunnel experiment** to evaluate **leading-edge extension (LEX) flow behavior** across multiple aircraft configurations at representative landing angles of attack.
- Developed **oil-flow visualization mixtures** and surface preparation techniques to capture separation lines, vortex formation, and reattachment regions.
- Processed and analyzed test imagery to extract **qualitative flow trends**, informing aerodynamic design decisions and configuration trade studies.

Technical Skills

Programming:
- Python
- MATLAB
- Mathematica

CAD & Drawings:
- SolidWorks
- NX CAD (Teamcenter Familiar)
- GD&T Fundamentals

Build & Manufacturing:
- Tooling & Fixture Design
- Hand Tools & Assembly
- 3D Printing & Composites